

A Poisson-Type Gravity Program from Organized Swirl Transport in Swirl–String Theory

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Abstract

This note develops a gravity program for Swirl–String Theory (SST) based on a single organizing principle: gravity is not identified with local swirl magnitude alone, but with the medium’s response to *organized swirl transport*. The derivation proceeds in full detail from incompressible, inviscid continuum mechanics in cylindrical geometry. We first define the source-string and the large inverse return-flow envelope, derive the full azimuthal velocity field, derive the pressure profile from steady Euler balance, and show explicitly that the naive identification

$$\mathbf{g} \propto -\nabla(\rho_f u^2)$$

fails to reproduce a Newtonian far field in the simplest single-string cylindrical model. This motivates a derived-field program in which a gravity potential Φ_{grav} is obtained from a Poisson equation whose source is built from organized transport. We then derive a tensorial source law

$$\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_i u_j),$$

its aligned-transport reduction,

$$\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_s^2 \hat{t}_i \hat{t}_j),$$

and practical axisymmetric reductions suitable for numerical implementation. All dimensional checks are performed explicitly, and the source-strength scales are evaluated using the canonical SST constants

$$\mathbf{v}_0 = 1.09384563 \times 10^6 \text{ m s}^{-1}, \quad r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad \rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}.$$

The result is a concrete and falsifiable gravity program: organized swirl transport defines the source sector; the gravity field is the long-range scalar response obtained by solving a Poisson problem sourced by transport organization.

Keywords: Poisson equation, organized swirl transport, gravity, Swirl–String Theory

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1 Purpose and status of this note

The purpose of this note is to formulate, in a mathematically explicit way, a gravity program for SST under the working postulate

$$\boxed{\text{Gravity is the macroscopic manifestation of organized swirl transport in the medium.}} \quad (1)$$

This note is *not* a proof that the program is correct. It is a derivation of the program itself, together with the reasons for each step and the diagnostic checks that motivate the final Poisson-type field equation.

The note is built around three claims.

Claim 1: A single source string embedded in an incompressible, inviscid cylindrical background naturally induces a return-flow organization consisting of a strong core and a weak, large-scale inverse envelope.

Claim 2: The primitive local transport-energy/stress scalar

$$\mathcal{I}_{\text{org}} \sim \rho_f u^2 \quad (2)$$

has the correct ontology—it belongs to the organized-transport sector—but if it is identified directly with the gravity field, the simplest cylindrical model yields the wrong far-field radial scaling.

Claim 3: A better formulation is obtained when organized transport acts as a *source* for a derived scalar field Φ_{grav} satisfying a Poisson equation. Outside the compact organized source region, the gravity field then solves the vacuum Laplace equation and can recover Newtonian-type $1/r^2$ behavior in the far field.

2 Canonical SST constants and dimensional baselines

We adopt the canonical SST constants:

$$\mathbf{v}_{\text{O}} = 1.09384563 \times 10^6 \text{ m s}^{-1}, \quad (3)$$

$$r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad (4)$$

$$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}. \quad (5)$$

The associated circulation-scale quantity is

$$A := \mathbf{v}_{\text{O}} r_c. \quad (6)$$

Numerically,

$$A = (1.09384563 \times 10^6)(1.40897017 \times 10^{-15}) \approx 1.54119586 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}. \quad (7)$$

The natural organized-transport stress scale is

$$\Sigma_0 := \rho_f \mathbf{v}_{\text{O}}^2. \quad (8)$$

Numerically,

$$\mathbf{v}_0^2 = (1.09384563 \times 10^6)^2 \approx 1.19649826 \times 10^{12} \text{ m}^2 \text{ s}^{-2}, \quad (9)$$

$$\Sigma_0 = (7.0 \times 10^{-7})(1.19649826 \times 10^{12}) \approx 8.37548784 \times 10^5 \text{ Pa}. \quad (10)$$

This quantity will recur throughout the note. It sets the characteristic transport-stress scale of the source string.

3 Kinematic decomposition and the meaning of organized transport

Let $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_s$ be a local adapted frame, where s denotes the local swirl-aligned or string-aligned direction. Then the velocity field is decomposed as

$$\mathbf{u} = u_r \hat{\mathbf{e}}_r + u_\theta \hat{\mathbf{e}}_\theta + u_s \hat{\mathbf{e}}_s. \quad (11)$$

The *swirl-aligned transport field* is the component tangent to the local organizing direction:

$$u_s = \mathbf{u} \cdot \hat{\mathbf{t}}, \quad \mathbf{u}_\parallel = u_s \hat{\mathbf{t}}, \quad (12)$$

where $\hat{\mathbf{t}}$ is the unit tangent to the local swirl-string / foliation direction.

This note will use two related transport objects:

Vector transport density

$$\mathbf{J}_{\text{org}} := \rho_f \mathbf{u}_\parallel = \rho_f u_s \hat{\mathbf{t}}, \quad (13)$$

with units

$$[\mathbf{J}_{\text{org}}] = \text{kg m}^{-2} \text{ s}^{-1}.$$

Organized transport tensor

$$\Sigma_{ij} := \rho_f u_i u_j. \quad (14)$$

This has units

$$[\Sigma_{ij}] = (\text{kg m}^{-3})(\text{m s}^{-1})(\text{m s}^{-1}) = \text{kg m}^{-1} \text{ s}^{-2} = \text{Pa}.$$

Thus Σ_{ij} is a stress-like object built from coherent transport.

If the dominant organized transport is aligned with the local tangent $\hat{\mathbf{t}}$, then

$$\Sigma_{ij}^{(\parallel)} = \rho_f u_s^2 \hat{t}_i \hat{t}_j. \quad (15)$$

4 The transport-asymmetry motivation

The conceptual motivation for distinguishing hidden aligned transport from observed transverse transport is the general structural lesson that, in organized media, the visible asymmetry can be controlled more strongly by transport redistribution and projection than by an explicit direct drift term. In the tokamak edge-transport study that motivated this line of thought, the projected parallel-flow contribution dominates the observable target flux in the simulated cases, and rotation plus drifts combine non-additively through momentum redistribution. This note does *not* identify SST with tokamak physics; it imports only the mathematical lesson that one

should separate the hidden aligned transport channel from the observed transverse response channel.

Accordingly, one can write a generic transport-response decomposition

$$\Gamma_\theta^{\text{SST}} = \rho_f \kappa u_s + \rho_f u_\theta^{(g)} - \partial_\theta \Pi + \rho_f D_\Omega \partial_\theta \Omega, \quad (16)$$

where:

- κ is a dimensionless projection factor,
- $u_\theta^{(g)}$ is a direct geometric transverse transport channel,
- Π is a stress/potential-like field,
- $D_\Omega \partial_\theta \Omega$ is an optional coarse-grained vorticity-transport closure.

This decomposition is not yet the gravity law. It is only the transport-theoretic motivation for taking Σ_{ij} or one of its derived scalars as the source-sector object.

5 Single source string in a large inverse return-flow envelope

We now derive the simplest incompressible, inviscid cylindrical model of a source string embedded in a large inverse return-flow background.

5.1 Source-string core

We model the source string as a Rankine-type swirl core:

$$u_\theta^{(\text{src})}(r) = \begin{cases} \frac{\mathbf{v}_\odot}{r_c} r, & 0 \leq r \leq r_c, \\ \frac{A}{r}, & r \geq r_c, \end{cases} \quad (17)$$

with

$$A = \mathbf{v}_\odot r_c.$$

This choice ensures continuity at $r = r_c$:

$$u_\theta^{(\text{src})}(r_c^-) = \mathbf{v}_\odot, \quad u_\theta^{(\text{src})}(r_c^+) = \frac{A}{r_c} = \mathbf{v}_\odot.$$

5.2 Inverse return envelope

We regularize the inverse background return flow by introducing a large envelope scale a . The background is taken to be opposite in sense to the source string:

$$u_\theta^{(\text{bg})}(r) = \begin{cases} -\eta \frac{A}{a^2} r, & 0 \leq r \leq a, \\ -\eta \frac{A}{r}, & r \geq a, \end{cases} \quad (18)$$

where η is the ratio of return-flow circulation to source-string circulation.

The total azimuthal velocity is

$$u_\theta(r) = u_\theta^{(\text{src})}(r) + u_\theta^{(\text{bg})}(r). \quad (19)$$

Therefore:

$$u_\theta(r) = \begin{cases} \left(\frac{\mathbf{v}_\odot}{r_c} - \eta \frac{A}{a^2} \right) r, & 0 \leq r \leq r_c, \\ A \left(\frac{1}{r} - \eta \frac{r}{a^2} \right), & r_c \leq r \leq a, \\ (1 - \eta) \frac{A}{r}, & r \geq a. \end{cases} \quad (20)$$

5.3 Cancellation radius

Inside the annular region $r_c < r < a$, the total swirl vanishes when

$$\frac{1}{r} - \eta \frac{r}{a^2} = 0.$$

Multiplying by r gives

$$1 - \eta \frac{r^2}{a^2} = 0,$$

hence

$$\boxed{r_0 = \frac{a}{\sqrt{\eta}}}. \quad (21)$$

This implies:

- if $\eta < 1$, the source-string sense dominates throughout the annulus;
- if $\eta = 1$, the flow reaches zero at $r = a$;
- if $\eta > 1$, the inverse background dominates beyond r_0 .

6 Pressure field from steady incompressible Euler balance

6.1 Radial balance

For steady, incompressible, axisymmetric swirl with no radial through-flow, the radial Euler balance is

$$\frac{dp}{dr} = \rho_f \frac{u_\theta^2}{r}. \quad (22)$$

This is the standard pressure support relation for steady swirl.

Dimensional check:

$$\left[\frac{dp}{dr} \right] = \text{Pa m}^{-1} = \frac{\text{kg}}{\text{m}^2 \text{s}^2},$$

while

$$\left[\rho_f \frac{u_\theta^2}{r} \right] = \frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}^2}{\text{s}^2} \cdot \frac{1}{\text{m}} = \frac{\text{kg}}{\text{m}^2 \text{s}^2}.$$

So (22) is dimensionally correct.

6.2 Source core pressure depression

Inside the source core, $u_\theta = (\mathbf{v}_\odot / r_c) r$. Therefore

$$\frac{dp}{dr} = \rho_f \frac{1}{r} \left(\frac{\mathbf{v}_\odot}{r_c} r \right)^2 = \rho_f \frac{\mathbf{v}_\odot^2}{r_c^2} r. \quad (23)$$

Integrating from $r = 0$ to $r = r_c$,

$$p(r_c) - p(0) = \int_0^{r_c} \rho_f \frac{\mathbf{v}_\odot^2}{r_c^2} r dr = \rho_f \frac{\mathbf{v}_\odot^2}{r_c^2} \cdot \frac{r_c^2}{2} = \frac{1}{2} \rho_f \mathbf{v}_\odot^2. \quad (24)$$

Hence the center-to-edge pressure depth is

$$\boxed{\Delta p_{\text{src}} = \frac{1}{2} \rho_f \mathbf{v}_\odot^2}. \quad (25)$$

Numerically,

$$\Delta p_{\text{src}} = \frac{1}{2} (7.0 \times 10^{-7}) (1.09384563 \times 10^6)^2 \approx 4.18774392 \times 10^5 \text{ Pa}. \quad (26)$$

6.3 Intermediate annulus

In the annulus $r_c < r < a$, from (20),

$$u_\theta(r) = A \left(\frac{1}{r} - \eta \frac{r}{a^2} \right).$$

Therefore

$$u_\theta^2 = A^2 \left(\frac{1}{r^2} - 2\eta \frac{1}{a^2} + \eta^2 \frac{r^2}{a^4} \right). \quad (27)$$

Substituting into (22),

$$\frac{dp}{dr} = \rho_f \frac{1}{r} A^2 \left(\frac{1}{r^2} - 2\eta \frac{1}{a^2} + \eta^2 \frac{r^2}{a^4} \right) \quad (28)$$

$$= \rho_f A^2 \left(\frac{1}{r^3} - \frac{2\eta}{a^2 r} + \eta^2 \frac{r}{a^4} \right). \quad (29)$$

Integrating term by term:

$$\int \frac{1}{r^3} dr = -\frac{1}{2r^2}, \quad (30)$$

$$\int \frac{1}{r} dr = \ln r, \quad (31)$$

$$\int r dr = \frac{r^2}{2}. \quad (32)$$

Thus the pressure difference between r_c and some $r \in (r_c, a)$ is

$$p(r) - p(r_c) = \rho_f A^2 \left[-\frac{1}{2r^2} + \frac{1}{2r_c^2} - \frac{2\eta}{a^2} \ln \left(\frac{r}{r_c} \right) + \frac{\eta^2}{2a^4} (r^2 - r_c^2) \right]. \quad (33)$$

6.4 Exactly screened case $\eta = 1$

When $\eta = 1$, the total velocity vanishes at $r = a$, and the pressure difference from the center to $r = a$ is

$$p(a) - p(0) = [p(r_c) - p(0)] + [p(a) - p(r_c)]. \quad (34)$$

Using (25) and (33) with $r = a$ and $\eta = 1$,

$$p(a) - p(0) = \frac{1}{2}\rho_f \mathbf{v}_\mathcal{O}^2 + \rho_f A^2 \left[-\frac{1}{2a^2} + \frac{1}{2r_c^2} - \frac{2}{a^2} \ln\left(\frac{a}{r_c}\right) + \frac{1}{2a^2} - \frac{r_c^2}{2a^4} \right]. \quad (35)$$

The $\pm 1/(2a^2)$ terms cancel, leaving

$$p(a) - p(0) = \frac{1}{2}\rho_f \mathbf{v}_\mathcal{O}^2 + \rho_f A^2 \left[\frac{1}{2r_c^2} - \frac{2}{a^2} \ln\left(\frac{a}{r_c}\right) - \frac{r_c^2}{2a^4} \right]. \quad (36)$$

Since $A = \mathbf{v}_\mathcal{O} r_c$,

$$\frac{A^2}{2r_c^2} = \frac{\mathbf{v}_\mathcal{O}^2}{2}.$$

Hence

$$p(a) - p(0) = \rho_f \mathbf{v}_\mathcal{O}^2 - 2\rho_f \frac{A^2}{a^2} \ln\left(\frac{a}{r_c}\right) - \frac{1}{2}\rho_f \frac{A^2 r_c^2}{a^4}. \quad (37)$$

For $a \gg r_c$, the last two terms are negligible, so

$$\boxed{p(a) - p(0) \approx \rho_f \mathbf{v}_\mathcal{O}^2.} \quad (38)$$

Numerically,

$$p(a) - p(0) \approx 8.37548784 \times 10^5 \text{ Pa} \quad (a \gg r_c, \eta = 1). \quad (39)$$

7 Why the naive gravity identification fails

A tempting first hypothesis is

$$\Phi_{\text{grav}}^{(0)} \propto \rho_f u_\theta^2, \quad \mathbf{g}^{(0)} \propto -\nabla(\rho_f u_\theta^2). \quad (40)$$

This is attractive because $\rho_f u_\theta^2$ is the primitive transport-stress scale.

However, in the intermediate source-dominated region $r_c \ll r \ll a$, one has

$$u_\theta(r) \approx \frac{A}{r}, \quad (41)$$

hence

$$\rho_f u_\theta^2 \approx \rho_f \frac{A^2}{r^2}. \quad (42)$$

Therefore

$$\frac{d}{dr}(\rho_f u_\theta^2) = \frac{d}{dr}\left(\rho_f \frac{A^2}{r^2}\right) = -2\rho_f \frac{A^2}{r^3}. \quad (43)$$

So the associated radial acceleration would scale as

$$g_r^{(0)} \propto \frac{1}{r^3}, \quad (44)$$

not as $1/r^2$.

This mismatch is decisive. It means that $\rho_f u^2$ has the right ontology but the wrong status:

$$\boxed{\text{organized transport should source gravity, but should not be identified directly with the gravity field.}} \quad (45)$$

8 The Poisson idea: organized transport as the source sector

The standard strategy in field theory and potential theory is to separate:

- a *source density*,
- a *propagated response field*.

So we introduce a scalar gravity potential Φ_{grav} satisfying

$$\boxed{\nabla^2 \Phi_{\text{grav}} = \mathcal{S}_{\text{grav}}}, \quad (46)$$

with

$$\boxed{\mathbf{g}_{\text{SST}} = -\nabla \Phi_{\text{grav}}}. \quad (47)$$

The question becomes: what scalar source $\mathcal{S}_{\text{grav}}$ should be built from organized transport?

8.1 Candidate scalar sources

Three natural candidates are:

Candidate A: transport-energy density

$$\mathcal{S}_A = \lambda_A \rho_f u_s^2. \quad (48)$$

Candidate B: divergence of transport vector

$$\mathcal{S}_B = \lambda_B \nabla \cdot (\rho_f u_s \hat{\mathbf{t}}). \quad (49)$$

Candidate C: curvature / convergence of transport stress

$$\mathcal{S}_C = \lambda_C \partial_i \partial_j \Sigma_{ij}, \quad \Sigma_{ij} = \rho_f u_i u_j. \quad (50)$$

Candidate A is the simplest, but tends to be too diffuse if used globally. Candidate B detects defects and source regions but may vanish for perfectly closed stationary loops. Candidate C measures the spatial organization of transport stress itself and is therefore the most structural.

Candidate C is therefore retained as the most structural stress diagnostic, but it is not elevated to an independent bulk monopole law unless supplemented by a genuine scalar density, an explicit boundary source, or a singular-core contribution.

9 Tensorial Poisson law for SST gravity

9.1 Definition of the source

Define the organized transport tensor

$$\Sigma_{ij} = \rho_f u_i u_j. \quad (51)$$

Then define the scalar source density

$$\boxed{\mathcal{S}_{\text{grav}} = \lambda \partial_i \partial_j \Sigma_{ij}}. \quad (52)$$

The gravity potential is then defined by

$$\boxed{\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_i u_j)}. \quad (53)$$

9.2 Dimensional analysis

The left-hand side has units

$$[\nabla^2 \Phi_{\text{grav}}] = [\Phi_{\text{grav}}] \text{m}^{-2}.$$

If Φ_{grav} is a gravitational potential, then

$$[\Phi_{\text{grav}}] = \text{m}^2 \text{s}^{-2},$$

so

$$[\nabla^2 \Phi_{\text{grav}}] = \text{s}^{-2}.$$

The tensor has units

$$[\Sigma_{ij}] = \text{Pa} = \text{kg m}^{-1} \text{s}^{-2}.$$

Applying two spatial derivatives gives

$$[\partial_i \partial_j \Sigma_{ij}] = \text{kg m}^{-3} \text{s}^{-2}.$$

Therefore λ must have units

$$[\lambda] = \text{m}^3 \text{kg}^{-1}. \quad (54)$$

This is closely analogous to a gravitational coupling scale.

9.3 Interpretation

Equation (53) says that gravity is sourced not by motion alone, and not by static pressure alone, but by the *spatial organization of coherent transport stress*. The double divergence extracts where coherent transport bends, converges, terminates, or changes structure.

This is the corrected research-track statement of the local stress-response program:

$$\boxed{\text{Stress organization supplies a local diagnostic; Newtonian gravity requires a monopole-bearing density or bou}} \quad (55)$$

Critical note: pressure equivalence and no bulk monopole. For constant-density incompressible Euler flow without external body force,

$$\partial_i \partial_j (\rho_f u_i u_j) = -\nabla^2 p. \quad (56)$$

Consequently the tensorial Candidate C is pressure-equivalent:

$$\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_i u_j) \implies \nabla^2 (\Phi_{\text{grav}} + \lambda p) = 0. \quad (57)$$

With decaying exterior boundary conditions this gives

$$\Phi_{\text{grav}} = -\lambda(p - p_\infty). \quad (58)$$

For smooth compact stress fields the integrated source $\int \partial_i \partial_j \Sigma_{ij} dV$ vanishes by Gauss' theorem, and the dipole moment also vanishes by integration by parts. The first possible far-field moment

is quadrupolar. Candidate C should therefore be retained as a local pressure/stress diagnostic unless an explicit boundary source or a non-divergence scalar density source is added. This is the same mathematical obstruction that appears in Lighthill's quadrupole source structure for aerodynamic sound [6].

10 Aligned-channel reduction

If the dominant transport channel is tangent to the local string direction $\hat{\mathbf{t}}$, then

$$u_i \approx u_s \hat{t}_i,$$

and therefore

$$\Sigma_{ij} \approx \rho_f u_s^2 \hat{t}_i \hat{t}_j. \quad (59)$$

Substituting into (53),

$$\boxed{\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j \left(\rho_f u_s^2 \hat{t}_i \hat{t}_j \right)}. \quad (60)$$

This is often the clearest SST form, because it makes explicit that:

- u_s^2 measures the strength of aligned organized transport,
- $\hat{t}_i \hat{t}_j$ carries the transport directionality,
- the double derivative measures how that organization varies in space.

11 Connection with transport asymmetry variables

The organized-transport viewpoint can also be expressed through the earlier azimuthal transport flux

$$\Gamma_\theta^{\text{SST}} = \rho_f \kappa u_s + \rho_f u_\theta^{(g)} - \partial_\theta \Pi + \rho_f D_\Omega \partial_\theta \Omega. \quad (61)$$

If one wants a reduced scalar source based directly on the observed projected transport channel, then a derived alternative is

$$\mathcal{S}_{\text{grav}}^{(\Gamma)} = \lambda_\Gamma \nabla \cdot (\Gamma_\theta^{\text{SST}} \hat{\mathbf{e}}_\theta), \quad (62)$$

or a second-derivative version,

$$\mathcal{S}_{\text{grav}}^{(\Gamma,2)} = \lambda_{\Gamma,2} \nabla^2 \Gamma_\theta^{\text{SST}}. \quad (63)$$

These are useful reduced diagnostics, but the tensor law (53) is more primitive and more covariant. Thus the hierarchy is:

$$\boxed{\Sigma_{ij} \text{ primitive source-sector object,} \quad \Gamma_\theta^{\text{SST}} \text{ observable projected transport channel.}} \quad (64)$$

12 Axisymmetric practical reduction

For numerical work, one often wants a scalar source that can be computed directly from axisymmetric fields. If the dominant organized transport is azimuthal,

$$u_\theta = u_\theta(r, z),$$

then the simplest scalar proxy is

$$\mathcal{U}(r, z) := \rho_f u_\theta^2(r, z). \quad (65)$$

A practical axisymmetric source density is then

$$\rho_{\text{grav}}^{(\text{SST})}(r, z) := \alpha_{\text{grav}} \left[\frac{\partial^2 \mathcal{U}}{\partial r^2} + \frac{1}{r} \frac{\partial \mathcal{U}}{\partial r} + \frac{\partial^2 \mathcal{U}}{\partial z^2} \right]. \quad (66)$$

This is simply

$$\rho_{\text{grav}}^{(\text{SST})} = \alpha_{\text{grav}} \nabla^2 \mathcal{U} \quad (67)$$

in axisymmetric cylindrical coordinates, ignoring θ -dependence.

One then solves

$$\nabla^2 \Phi_{\text{grav}} = 4\pi G_{\text{eff}} \rho_{\text{grav}}^{(\text{SST})}. \quad (68)$$

Dimensional note on α_{grav} . In the practical axisymmetric proxy

$$\rho_{\text{grav}}^{(\text{SST})} = \alpha_{\text{grav}} \nabla^2 (\rho_f u_\theta^2), \quad (69)$$

α_{grav} is not dimensionless. Since

$$[\rho_{\text{grav}}^{(\text{SST})}] = \text{kg m}^{-3}, \quad [\nabla^2 (\rho_f u_\theta^2)] = \text{kg m}^{-3} \text{s}^{-2}, \quad (70)$$

one must have

$$[\alpha_{\text{grav}}] = \text{s}^2. \quad (71)$$

Equivalently, in $\nabla^2 \Phi_{\text{grav}} = 4\pi G_{\text{eff}} \alpha_{\text{grav}} \nabla^2 (\rho_f u_\theta^2)$, α_{grav} supplies the missing time-scale squared. A natural but not yet derived candidate is

$$\alpha_{\text{grav}} \sim \left(\frac{r_c}{c} \right)^2 \approx 1.6595 \times 10^{-42} \text{s}^2, \quad (72)$$

which is **[SPECULATIVE]** until obtained from a closure theorem or calibration.

Important remark. Equation (68) is *not* identical to

$$\Phi_{\text{grav}} \propto \mathcal{U}.$$

Instead, $\mathcal{U}$ produces a compact source density $\rho_{\text{grav}}^{(\text{SST})}$, and Φ_{grav} is obtained by solving the Poisson problem with that source.

13 Vacuum region and Newtonian far-field recovery

Suppose the organized transport source occupies a compact region V_{src} , and outside that region the source vanishes:

$$\mathcal{S}_{\text{grav}} = 0 \quad \text{for } \mathbf{x} \notin V_{\text{src}}. \quad (73)$$

Then outside the source region,

$$\nabla^2 \Phi_{\text{grav}} = 0. \quad (74)$$

In spherical symmetry, the general decaying solution is

$$\Phi_{\text{grav}}(r) = -\frac{K}{r}, \quad (75)$$

so

$$g_r(r) = -\frac{d\Phi_{\text{grav}}}{dr} = -\frac{K}{r^2}. \quad (76)$$

Thus:

Once the source is compact, the Poisson program naturally allows a $1/r^2$ far field.

(77)

This is the main reason to prefer the Poisson program over the direct identification $g \sim -\nabla(\rho_f u^2)$.

14 Effective gravitational mass from organized transport

If the potential obeys

$$\nabla^2 \Phi_{\text{grav}} = \mathcal{S}_{\text{grav}},$$

then integrating over volume gives

$$\int_V \nabla^2 \Phi_{\text{grav}} dV = \int_V \mathcal{S}_{\text{grav}} dV. \quad (78)$$

By Gauss' theorem,

$$\oint_{\partial V} \nabla \Phi_{\text{grav}} \cdot d\mathbf{S} = \int_V \mathcal{S}_{\text{grav}} dV. \quad (79)$$

For a localized source and asymptotic field

$$\Phi_{\text{grav}} \sim -\frac{G_{\text{eff}} M_{\text{grav}}^{(\text{SST})}}{r},$$

one obtains

$$\oint_{\partial V} \nabla \Phi_{\text{grav}} \cdot d\mathbf{S} = 4\pi G_{\text{eff}} M_{\text{grav}}^{(\text{SST})}. \quad (80)$$

Hence

$M_{\text{grav}}^{(\text{SST})} = \frac{1}{4\pi G_{\text{eff}}} \int_V \mathcal{S}_{\text{grav}} dV.$

(81)

For the tensorial source law (53),

$M_{\text{grav}}^{(\text{SST})} = \frac{\lambda}{4\pi G_{\text{eff}}} \int_V \partial_i \partial_j (\rho_f u_i u_j) dV.$

(82)

For smooth compact incompressible stress fields this integral vanishes by the no-monopole result above. Equation (82) should therefore be read as a boundary/singular-core diagnostic, not as a closed bulk mass law. A nonzero Newtonian monopole requires a genuine scalar density source such as $\rho_m = \rho_E/c^2$, or an explicitly modeled boundary source.

15 Scaling estimate for the source strength

Take the organized transport scale

$$\Sigma_0 = \rho_f \mathbf{v}_G^2 \approx 8.38 \times 10^5 \text{ Pa}.$$

If the source varies over a characteristic length L , then

$$\partial_i \partial_j \Sigma_{ij} \sim \frac{\Sigma_0}{L^2}. \quad (83)$$

Thus the source-density scale is

$$\mathcal{S}_{\text{grav}} \sim \lambda \frac{\Sigma_0}{L^2}. \quad (84)$$

If one takes $L = r_c$, then

$$\frac{\Sigma_0}{r_c^2} = \frac{8.37548784 \times 10^5}{(1.40897017 \times 10^{-15})^2} \quad (85)$$

$$\approx 4.22 \times 10^{35} \text{ Pa m}^{-2}. \quad (86)$$

This is enormous. Therefore the coupling λ must be very small if ordinary gravitational strengths are to emerge from such a source. This is not a defect of the program; it implies that the organized transport reservoir is strong and gravity is a weak response channel.

16 Comparison of source choices

For clarity, we collect the main source candidates:

$$\mathcal{S}_A = \lambda_A \rho_f u_s^2, \quad (87)$$

$$\mathcal{S}_B = \lambda_B \nabla \cdot (\rho_f u_s \hat{\mathbf{t}}), \quad (88)$$

$$\mathcal{S}_C = \lambda_C \partial_i \partial_j (\rho_f u_i u_j). \quad (89)$$

Why $\mathcal{S}_A$ is insufficient as the main law. It is the correct primitive stress scale, but when interpreted too literally it leads to the wrong radial scaling in the simplest cylindrical model.

Why $\mathcal{S}_B$ is useful but incomplete. It detects source/sink/convergence structure but may vanish for perfectly closed transport without defects.

Why $\mathcal{S}_C$ is retained only as a diagnostic. It depends on the spatial organization of coherent transport stress itself and naturally distinguishes trivial transport from transport whose geometry biases the surrounding medium. However, for smooth compact incompressible fields it is a double-divergence source with zero monopole. It is therefore preferred only as a local stress/pressure diagnostic or boundary-source detector, not as the primary Newtonian mass law.

17 A practical simulation workflow

The stress-diagnostic and density-source gravity program can be implemented numerically as follows. The two branches must be tested separately so that the local stress channel is not confused with the monopole-bearing Poisson channel.

Step 1: compute the flow field

Obtain the full velocity field $\mathbf{u}(\mathbf{x})$ from the SST simulation.

Step 2: compute the organized transport tensor

$$\Sigma_{ij}(\mathbf{x}) = \rho_f u_i(\mathbf{x}) u_j(\mathbf{x}). \quad (90)$$

Step 3: build the source density

Either compute the full tensorial diagnostic

$$\mathcal{S}_{\text{stress}}(\mathbf{x}) = \lambda \partial_i \partial_j \Sigma_{ij}(\mathbf{x}), \quad (91)$$

and verify its no-monopole behaviour for compact smooth fields, or compute a separate monopole-bearing density source. For axisymmetric stress diagnostics one may use

$$\rho_{\text{grav}}^{(\text{SST})}(r, z) = \alpha_{\text{grav}} \nabla^2(\rho_f u_\theta^2). \quad (92)$$

Step 4: solve the Poisson problem

$$\nabla^2 \Phi_{\text{grav}} = \mathcal{S}_{\text{grav}}. \quad (93)$$

Step 5: compute the gravity field

$$\mathbf{g}_{\text{SST}} = -\nabla \Phi_{\text{grav}}. \quad (94)$$

Step 6: compare against alternative predictors

To test whether gravity is indeed transport-based, compare the predictive power of:

$$\Omega, \quad \rho_f u^2, \quad \Gamma_\theta^{\text{SST}}, \quad \partial_i \partial_j (\rho_f u_i u_j), \quad \Phi_{\text{grav}}. \quad (95)$$

18 Falsifiability and diagnostic tests

The program is falsifiable. At minimum, the following tests should be applied.

18.1 Test 1: source compactness

The tensorial source

$$\partial_i \partial_j (\rho_f u_i u_j)$$

should be localized around organized source structures, not diffuse across the whole domain. If it is diffuse and non-decaying everywhere, the Poisson picture loses its Newtonian-like far-field advantage.

18.2 Test 2: far-field scaling

For isolated organized sources, the solved field Φ_{grav} should approach a harmonic vacuum solution outside the source region, and $\mathbf{g}_{\text{SST}}$ should approach $1/r^2$ -type scaling rather than $1/r^3$.

18.3 Test 3: topology sensitivity

If gravity is sourced by organized transport, then two configurations with comparable local swirl magnitude but different transport organization should generate measurably different Φ_{grav} .

18.4 Test 4: defect sensitivity

If the source depends on curvature/convergence of transport organization, then topological reconnections or strong curvature concentrations should leave signatures in $\mathcal{S}_{\text{grav}}$.

18.5 Test 5: comparison with Swirl Clock

If Swirl Clock is also transport-emergent, then Φ_{grav} and $S_{(t)}^o$ should correlate more strongly with organized-transport diagnostics than with Ω alone.

19 Assumptions, caveats, and status tags

19.1 Assumptions

- (A1) The medium is incompressible and inviscid at the fundamental level.
- (A2) Organized transport is the ontological source sector for gravity.
- (A3) A scalar gravity field is sufficient as the first derived response field.
- (A4) The source region is effectively compact or can be coarse-grained into a compact source.
- (A5) The organized transport tensor $\Sigma_{ij} = \rho_f u_i u_j$ is the correct primitive continuum object.

19.2 Caveats

- (C1) The tensorial source law (53) is a proposal, not a theorem.
- (C2) The scalar coupling λ remains to be fixed by closure, matching, or experiment.
- (C3) The axisymmetric proxy (68) is computationally convenient but less invariant than the full tensor law.
- (C4) The single-source cylindrical model is only a diagnostic toy model; realistic knot/string structures are three-dimensional and topologically nontrivial.

19.3 Status

Status: speculative but mathematically well-posed and directly testable.	(96)
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20 Conclusion

We summarize the derivation.

1. A source string embedded in an incompressible inviscid background naturally generates a strong local organized transport core plus a weak large-scale inverse return-flow envelope.
2. The primitive transport-stress quantity

$$\rho_f u^2$$

has the correct ontological role but fails when identified directly with the gravity field, because the simplest cylindrical model gives a $1/r^3$ acceleration rather than a $1/r^2$ far field.

3. This failure is repaired by elevating organized transport to the *source sector* rather than the *field itself*.
4. The resulting gravity program is:

$$\boxed{\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_i u_j), \quad \mathbf{g}_{\text{SST}} = -\nabla \Phi_{\text{grav}}.} \quad (97)$$

5. In the aligned-channel reduction,

$$\boxed{\nabla^2 \Phi_{\text{grav}} = \lambda \partial_i \partial_j (\rho_f u_s^2 \hat{t}_i \hat{t}_j).} \quad (98)$$

6. In practical axisymmetric computations, one may use

$$\boxed{\nabla^2 \Phi_{\text{grav}} = 4\pi G_{\text{eff}} \alpha_{\text{grav}} \nabla^2 (\rho_f u_\theta^2)} \quad (99)$$

inside the source region, matched to Laplace vacuum outside.

Therefore the gravity program obtained in this note is:

$$\boxed{\text{organized swirl transport defines the source;} \quad \text{gravity is the Poisson-solved scalar response field.}} \quad (100)$$

A physically accurate child-level analogy is: the swirl itself is not yet the pull; the pull comes from how the swirl is organized in space, and the rest of the medium responds to that organized pattern.

A Numerical values collected in one place

$$\mathbf{v}_\text{O} = 1.09384563 \times 10^6 \text{ m s}^{-1}, \quad (101)$$

$$r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad (102)$$

$$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}, \quad (103)$$

$$A = \mathbf{v}_\text{O} r_c \approx 1.54119586 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}, \quad (104)$$

$$\Sigma_0 = \rho_f \mathbf{v}_\text{O}^2 \approx 8.37548784 \times 10^5 \text{ Pa}, \quad (105)$$

$$\Delta p_{\text{src}} = \frac{1}{2} \rho_f \mathbf{v}_\text{O}^2 \approx 4.18774392 \times 10^5 \text{ Pa}. \quad (106)$$

B Possible next derivation

The next logical derivation after this note is to compute the far-field multipole reduction of

$$\partial_i \partial_j (\rho_f u_i u_j),$$

for localized knot or loop sources and identify the conditions under which the monopole part is nonzero, suppressed, or topology-dependent.

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